f) $\exists n \exists m (n^2+m^2=6)$. Let it be True. By defn. of existential quantifier, there exists an integer a in the domain such that $\exists m (a^2+m^2=6)$ is True. By defn. of existential quantifier, there exists an integer b in the domain such that $(a^2+b^2=6)$ is True.
We Know, $a^2 \geq 0$, $b^2 \geq 0$. If $a^2 > 6$, then $b^2 < 0$ which is a contradiction.
$\therefore a^2 \leq 6$. Similarly $b^2 \leq 6$. $3^2 = 9 > 6$. $\therefore a, b < 3$.

$\boxed{\text{Case 1:}}$ $a=0, b=0$ $a^2+b^2=0 \neq 6$

$\boxed{\text{Case 2:}}$ $a=0, b=1$ and $a=1, b=0$ $a^2+b^2=1 \neq 6$

$\boxed{\text{Case 3:}}$ $a=1, b=1$ $a^2+b^2=2 \neq 6$ $\boxed{\text{Case 4:}}$ $a=1, b=2$ and $a=2, b=1$ $a^2+b^2=5 \neq 6$

$\boxed{\text{Case 5:}}$ $a=2, b=2$ $a^2+b^2=8 \neq 6$

$\therefore (a^2+b^2=6)$ is False. $\therefore \exists n \exists m (n^2+m^2=6)$ is False

g) $\exists n \exists m (n+m=4 \land n-m=1)$. Let it be True. By defn. of existential quantifier, there exists an integer a in the domain s.t. $\exists m (a+m=4 \land a-m=1)$ is True. By defn. of existential quantifier, there exists an integer b in the domain s.t. $(a+b=4 \land a-b=1)$ is True. By defn. of and, $a+b=4$ is True and $a-b=1$ is True. $(a+b)+(a-b)=4+1 \Rightarrow 2a=5 \Rightarrow a=2.5 \notin \mathbb{Z}$.
This is a contradiction. $\therefore \exists n \exists m (n+m=4 \land n-m=1)$ is False.

h) $\exists n \exists m (n+m=4 \land n-m=2)$.
Let, $x=3, y=1$. $\therefore x+y=4$ and $x-y=2$. By defn. of AND, $(x+y=4 \land x-y=2)$ is True. By defn. of existential quantifier, $\exists m (x+m=4 \land x-m=2)$ is True. By defn. of existential quantifier, $\exists n \exists m (n+m=4 \land n-m=2)$ is True.

i) $\forall n \forall m \exists p (p=(m+n)/2)$. Let it be True. By defn. of negation, $\neg \forall n \forall m \exists p (p=(m+n)/2)$ is False. By De Morgan's law of quantifiers, $\exists n \neg \forall m \exists p (p=(m+n)/2)$ is False. By De Morgan's law of quantifiers, $\exists n \exists m \neg \exists p (p=(m+n)/2)$ is False. By De Morgan's law of quantifiers, $\exists n \exists m \forall p \neg (p=(m+n)/2)$ is False.